MGSC 2301 · PROFESSOR DENISE SPENCER · EXAM 2

Business Statistics Exam 2

Chapters 6–8 in one place: the key ideas, formulas and course examples for each chapter. Formulas use the notation and labels from the *Exam 2: Chapters 6-8 Formulas* sheet.

Exam 2 at a glance

COVERAGE
Chapters 6, 7, 8

FORMAT
50 questions, 59 pts + 3 extra credit. Q1–12 multiple choice, Q13–50 fill-in-the-blank

TIME
95 minutes in class, Respondus LockDown Browser

ALLOWED
Calculator, z-table, t-table, formula sheet (no notes), blank paper

Show handwritten work on every fill-in question except the 13 marked exempt, put your name on every page, and hand it in. Enter every answer in Canvas.

CHAPTER 6

The Normal Distribution

Normal distribution
Continuous, bell-shaped and symmetric, described by μ and σ. Mean = median = mode. Total area under the curve = 1, with .50 on each side of μ.

Standard normal distribution
μ = 0 and σ = 1. A *z*-score converts any normal value to this scale: the number of standard deviations above (+) or below (–) the mean.

Empirical rule
About 68%, 95% and 99.7% of values fall within 1, 2 and 3σ of μ. For exactly 95%, use 1.96.

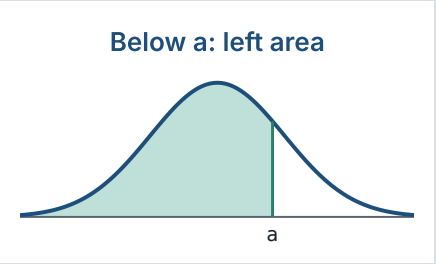
Continuous variable
P(X = a) = 0, so P(X < a) = P(X ≤ a).

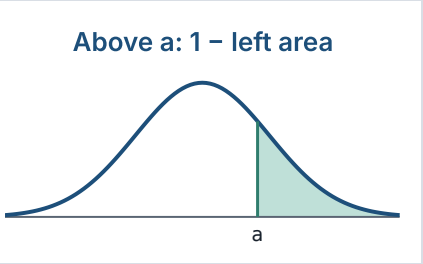
Formulas

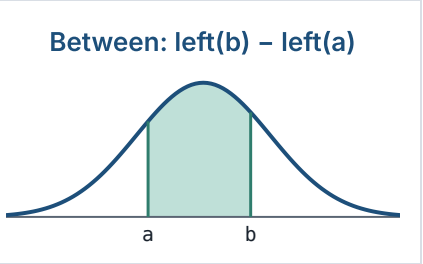
$z = \dfrac{x - \mu}{\sigma}$	*z*-score (raw data)
$\mathbf{x} = \mu + (\mathbf{z})(\sigma)$	*z*-score to a raw score
$\mu = \mathbf{x} - (\mathbf{z})(\sigma)$	*z*-score to a population mean

Finding a probability (value → area)

The course z-table is **cumulative: it gives the area to the LEFT of z**. Round z to 2 decimals, then:







Finding a value (area → value)

Turn the wording into a **left area** (top 10% → .90, bottom 20% → .20), find that area inside the table, read z, then use x = μ + (z)(σ).

> **Course example: goldfish lifetimes, μ = 60 days, σ = 20.** More than 80 days: z = 1.00, 1 − .8413 = **.1587**. Between 25 and 45: .2266 − .0401 = **.1865**. Top 10%: 60 + 1.28(20) = **85.6 days**.

CHAPTER 7

Sampling and Sampling Distributions

Parameter vs. statistic

A parameter describes the population (μ, σ, p). A statistic describes a sample ($\bar{x}$, s, $\bar{p}$).

Sampling distribution

The distribution of a statistic over all possible samples of the same size n. Its standard deviation is the **standard error**.

Sampling error

The natural difference between a statistic and the parameter from sample to sample. It isn't a mistake.

Random sampling

Simple random, stratified random (sample from every subgroup), systematic (every kth item, k = N/n), cluster (randomly pick some groups).

Nonrandom sampling

Convenience, judgment, quota.

Survey errors

Coverage (selection), nonresponse, measurement, and sampling error.

Sampling distribution of $\bar{x}$

$E(\bar{x}) = \mu_{\bar{X}} = \mu$	Expected value for sampling distribution of $\bar{x}$
$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$	Standard error of the mean (σ known)
$z_{\bar{x}} = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$	*z*-score (sample mean)

Shape: normal for any n if the population is normal. Otherwise approximately normal when **n ≥ 30** (Central Limit Theorem).

Sampling distribution of $\bar{p}$

$\bar{p} = \frac{x}{n}$	Sample proportion
$E(\bar{p}) = p.$	Expected value for sampling distribution of $\bar{p}$
$\mathrm{np} \geq 5, \text{ and } \mathrm{n(1-p)} \geq 5$	Normality requirements for sampling distribution of $\bar{p}$
$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$	Standard error of the proportion
$z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}}$	*z*-score (sample proportion)

Course examples. Credit cards, μ = $2,864, σ = $775, n = 50: $\sigma_{\bar{X}}$ = 109.6016, P($\bar{x}$ > $3,000) = **.1075**. Internet use, p = .66, n = 300: $\sigma_{\bar{p}}$ = .0273, P(within .04 of p) = **.8584**.

When asked to describe a sampling distribution, state all three: center (expected value), spread (standard error), and shape (normal or not, with the n ≥ 30 or np ≥ 5 check).

CHAPTER 8

Estimation for Single Populations

Point estimate
One statistic used to estimate a parameter: $\bar{x}$ for μ, $\bar{p}$ for p.

Interval estimate (confidence interval)
Point estimate ± margin of error. Width = 2E.

Critical probability
$\frac{\alpha}{2}$ & $1 - \frac{\alpha}{2}$, where α = 1 − confidence level. 95% → .0250 and .9750.

Critical value
The z or t that cuts off those areas: 90% → 1.645, 95% → 1.96, 98% → 2.33, 99% → 2.576.

t distribution
Used when σ is unknown. Bell-shaped with heavier tails than z. Degrees of freedom df = n − 1.

Minimum sample size
Smallest n that gives the desired margin of error E. Always round **up**.

Formulas

$\mu = \bar{x} \pm (Z_{\alpha/2}) \left(\frac{\sigma}{\sqrt{n}} \right)$	CI estimate for μ (σ known)
$S_{\bar{X}} = \frac{s}{\sqrt{n}}$	Standard error of the mean (σ unknown)
$\mu = \bar{x} \pm (t_{\alpha/2}) \left(\frac{s}{\sqrt{n}} \right)$	CI estimate for μ (σ unknown)
$p = \bar{p} \pm (Z_{\alpha/2}) \sqrt{\frac{\bar{p}(1-\bar{p})}{\mathrm{n}}}$	CI estimate for *p*
$\mathrm{n} = \left(\frac{(z_{\alpha/2})\sigma}{E} \right)^2$	Sample size for interval estimate of μ
$n = \frac{(z_{\alpha/2})^2 p^*(1-p^*)}{E^2}$	Sample size for interval estimate of *p*

- **z or t:** σ given → z. Only s given → t with df = n − 1, even when n is 30 or more.

- **Reading the t-table:** row = df, column = upper-tail area α/2. Example: df = 17, α/2 = .05 → t = 1.740.
- **p*:** the pilot-study estimate. With no pilot, use p* = .50.
- **Interpretation:** "We are 95% confident that the population mean (or proportion) is between L and U."

> **Course examples.** Verizon, $\bar{x}$ = 153, σ = 46, n = 85, 95%: **(143.22, 162.78)**. Aerospace managers, $\bar{x}$ = 13.56, s = 7.8, n = 18, 90%, t = 1.740: **(10.36, 16.76)**. Temporary hires, $\bar{p}$ = .39, n = 87, 95%: **(.2875, .4925)**. Acme, 98%, E = .03, p* = .40: n = 1447.71 → **1,448**.

Across chapters: what gets mixed up

THE QUESTION IS ABOUT…	USE
One value x (Ch 6)	$z = \dfrac{x - \mu}{\sigma}$
A sample mean $\bar{x}$ (Ch 7)	$z_{\bar{x}} = \dfrac{\bar{x} - \mu}{\sigma_{\bar{x}}}$, with $\sigma_{\bar{X}} = \dfrac{\sigma}{\sqrt{n}}$
A sample proportion $\bar{p}$ (Ch 7)	$z_{\bar{p}} = \dfrac{\bar{p} - p}{\sigma_{\bar{p}}}$, with the known p
Estimating μ or p (Ch 8)	Point estimate ± (critical value)(standard error)

- **p vs. $\bar{p}$:** Chapter 7 probabilities use the known p in the standard error. A Chapter 8 CI uses $\bar{p}$, because p is unknown.
- **"E" means two things:** E($\bar{x}$) is an expected value. E alone is the margin of error. "Within 3 percentage points" means E = .03.
- **Middle 95%:** of individual values is μ ± 1.96σ. A 95% CI is $\bar{x}$ ± 1.96σ/√n.
- **Calculator vs. table:** the answer key uses table values (z to 2 decimals, 4-place areas, 2.33 for 98%). If your calculator disagrees, enter the table answer.
- **Rounding:** follow "round final answer only" or "round all calculations" as each question says. Minimum n always rounds up.

Other sheets

More detail lives on the companion pages:

Z-Score Cheat Sheet | Exam 2 Calculator Sheet | EXAM2 Calculator Files